

List Recovery, Entropy Inequalities, and Combinatorial Designs

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1 Introduction

We are primarily concerned with the below conjecture [CZ25] :

Conjecture 1 (Chen-Zhang Conjecture). *For any constants $\varepsilon > 0$, $\ell \geq 2$, $L + 1 = \ell^a$, $a \in \mathbb{N}_{\geq 2}$, $R \leq \frac{a-1}{a}$, and generator γ of $\mathbb{F}_q^\times$, there exists a constant C such that: if $s \geq C$, $\frac{k-1}{s} \geq a$, and n is sufficiently large, then any rate $R = \frac{k}{sn}$ folded Reed-Solomon code*

$$\text{FRS}_{n,k}^{s,\gamma}(\alpha_1, \alpha_2, \dots, \alpha_n)$$

with appropriate evaluation points in $\mathbb{F}_q$ is $(\rho^ - \varepsilon, \ell, L)$ list-recoverable, where*

$$\rho^* = \frac{L+1-\ell}{L+1} \left(1 - \frac{aR}{a-1} \right).$$

It was previously proven [CZ25] that this conjectured value for ρ^* is, in fact, an *upper bound*.

More specifically, it was proven that it is impossible for a folded Reed-Solomon code to be $(\rho^* + o(1), \ell, L)$ list-recoverable if $\rho^* = \frac{L+1-\ell}{L+1} \left(1 - \frac{aR}{a-1} \right)$ and $a := \log_\ell(L+1) > 1$.

Then, obviously, if ρ is larger than the specification above, if we are less strict with regards to the criteria that a codeword must meet in order to be admitted into our output list, then the output list size will naturally grow; thus, no folded Reed-Solomon code can be (ρ, ℓ, L) list-recoverable for a ρ greater than the aforementioned above specification.

Theorem 1. [CZ25] *Let $s, n, L \geq 1$, $k \in [n]$, $2 \leq \ell \leq L$, $q \geq \ell$, generator γ of $\mathbb{F}_q^\times$, and*

$$a = \lceil \log_\ell(L+1) \rceil > 1.$$

Suppose $R \leq \frac{a-1}{a}$ and $\frac{k-1}{s} \geq a$. If a folded Reed-Solomon code

$$\text{FRS}_{n,k}^{(s,\gamma)}(\alpha_1, \alpha_2, \dots, \alpha_n)$$

of rate $R = \frac{k}{sn}$ with appropriate evaluation points in $\mathbb{F}_q$ is (ρ, ℓ, L) list-recoverable, then

$$\rho \leq \frac{L+1-\ell}{L+1} \left(1 - \frac{aR}{a-1} \right) + \frac{5}{n}.$$

Conjecture 1 (The Chen-Zhang Conjecture) is derived from the belief that this rigorously proven upper bound in Theorem 1.7 may actually be tight, that the lower bound might be incredibly close to this upper bound.

However, note that this is an *achievability* lower bound, as opposed to an *existence* lower bound. In other words, when the conjecture says that the code is $(\rho^* - \varepsilon, \ell, L)$ list-recoverable, this does not mean that it is impossible for the code to be list-recoverable for smaller values of ρ (assume ℓ and

L are unchanged). After all, decreasing the value of ρ means we are increasing the strictness of the criteria for deciding whether to allow a codeword into the output list. Thus, any code that is (ρ, ℓ, L) list-recoverable is also (ρ', ℓ, L) list-recoverable, where $\rho' < \rho$.

So, rather than being interested in a lower bound and upper bound pair for *allowable* ρ , we are interested in bounds for the list-recovery *performance limit* of folded Reed–Solomon codes. We want to fully understand exactly how high we can push the fraction of allowable errors, the error tolerance, of folded Reed–Solomon codes before output list size explodes.

Theorem 2. [CZ25] Let $n, L \geq 1$, $k \in [n]$, $q > n$, $s \geq 3$, and generator γ of $\mathbb{F}_q^\times$. The folded Reed–Solomon code

$$\text{FRS}_{n,k}^{(s,\gamma)}(\alpha_1, \alpha_2, \dots, \alpha_n)$$

with appropriate evaluation points in $\mathbb{F}_q$ is

$$\left(\frac{1}{2} - \frac{sR}{s-2}, 2, 3 \right) \text{ list-recoverable.}$$

This theorem tells us that the Chen-Zhang Conjecture holds true when $(\ell, L) = (2, 3)$. Their proof involves the use of geometric table-agreement hypergraphs. This is the smallest non-trivial list-recoverability case, as for smaller choices of the values in the (ℓ, L) pair, we can just use prior list-decoding results to prove tightness of the bound.

There still exists a massive gap between what has already been proven and what needs to be proven in order to confirm or reject this conjecture in totality. We seek to further push prior efforts to close this gap.

Very recent work [BCDZ25a] has helped to further close this gap, by proving yet another particular case of the Chen-Zhang Conjecture. Specifically, we are interested in the proof of the $\rho^* = 0$ case of the conjecture.

Obviously, the achievability result suggested by the conjecture is most useful and relevant from a theory perspective when it holds for any general tuple (ℓ, L, R) , when we can confidently assert the list-recoverability validity of this general formula for ρ^* that corresponds to any such tuple. In other words, ρ^* is a function of ℓ, L, R . Then, the $\rho^* = 0$ case consists of two subcases, as we are interested in the ℓ, L, R values for which the conjecture holds, and so we are interested in the ℓ, L, R values for which $\rho^* = 0$:

- $L + 1 = \ell$
- $R = \frac{\ell-1}{\ell}$; in other words, this subcase corresponds to when rate is maximized

Quick note: when $\rho^* = 0$, we can drop the ε term in the Chen-Zhang Conjecture, as error tolerance cannot drop below 0.

2 Motivations and Applications

It turns out that there is a “local equivalence between random linear codes and (nearly) optimal subspace design codes up to an arbitrarily small rate decrease.” [BCDZ25b] In other words, random linear codes and folded Reed–Solomon codes, as they belong to this new large category of codes (this will make much more sense once we properly introduce and define the notion of subspace design and how they play a role in codes), both behave similarly for any given local property, so long as their rates are arbitrarily close. We will introduce the definition of a local property soon.

You may also free to write some ideas from [RV25]’s section 5, which covers list recovery applications

3 Literature Review

“Explicit Folded Reed–Solomon and Multiplicity Codes Achieve Relaxed Generalized Singleton Bounds” [CZ25] proves a strict upper bound on the error tolerance of folded Reed–Solomon codes; Previously, it was conjectured that the upper bound was higher, but this result by Yeyuan Chen and Zihan Zhang proves an upper bound, above which folded Reed–Solomon codes fail to be list-recoverable, that is lower than the aforementioned old conjectured upper bound. Thus, this old conjecture (shown below) has since been refuted, but it is nonetheless worth understanding why it seemed reasonable at the time.

Conjecture 2 (Disproved Conjecture on List-Recovery Radius). [Gur11] [GW13] *Explicit folded Reed–Solomon codes are (ρ, ℓ, L) list-recoverable for radius*

$$\rho = \frac{1}{L+1} \left(L+1 - \ell - \frac{sLR}{s-L+1} \right).$$

More accurately, while Guruswami did not specify a particular upper bound, he instead suggested that list-recoverability held true *above* the upper bound which we now know to be true.

Below are some important landmark results.

[LMS25] proved that random Reed–Solomon codes and random linear codes are locally equivalent. In simpler terms, this just means that such codes look the same to any test that only checks relatively few entries. The result regarding local equivalence that we say in [BCDZ25b] is a more general result that applies to *all* (nearly) optimal subspace design codes.

[LS25] established the previously best known upper and lower bounds for smallest near optimal worst-case output list size for list recovery of random linear codes. Their proof builds off the techniques of [ZP81], but they then replace the counting argument in that paper with an argument about the maximum intersection of a “list-recovery ball” and a low-dimensional subspace with large distance, inspired by [RS].

[DMRR25]’s argument differs from [LS25] because it is less general. While [LS25]’s results are independent of alphabet size, [DMRR25] only works over small constant-sized alphabets/fields. They take [GHK10]’s arguments but extend them, using ideas about list-decoding and generalizing them to apply to list-recovery. The small alphabet regime is simply less understood and explored by researchers, and so this work gives much-needed attention to it. However, it being less general means that it is less immediately relevant to our research problem of interest.

The below table summarizes several recent results in list recovery research [BCDZ25a].

Codes	Alphabet size $ \Sigma $	List size L	Explicit?
Random Linear Codes [ZP81, Gur01]	$\ell^{O(1/\varepsilon)}$	$ \Sigma ^{O(\ell/\varepsilon)}$	No
Random Linear Codes [RW18]	$\ell^{O(1/\varepsilon)}$	$ \Sigma ^{O(\log^2(\ell)/\varepsilon)}$	No
Random Linear Codes [LS25]	$\ell^{O(1/\varepsilon)}$	$(\ell/\varepsilon)^{O(\ell/\varepsilon)}$	No
Random RS Codes [LS25]	$O_{\varepsilon,\ell,L,R}(n)$	$(\ell/\varepsilon)^{O(\ell/\varepsilon)}$	No
Random Linear Codes (This Work)	$\ell^{(\ell/(R+\varepsilon))^{O(R/\varepsilon)}/\varepsilon}$	$\left(\frac{\ell}{R+\varepsilon}\right)^{O(R/\varepsilon)}$	No
Random RS Codes (This Work)	$O_{\varepsilon,\ell,L,R}(n)$	$\left(\frac{\ell}{R+\varepsilon}\right)^{O(R/\varepsilon)}$	No
Folded RS Codes [GR06]	$\left(\frac{n}{\varepsilon^2}\right)^{O(\log(\ell)/\varepsilon^2)}$	$\left(\frac{n}{\varepsilon^2}\right)^{O(\log(\ell)/\varepsilon^2)}$	Yes
Univariate Multiplicity Codes [Kop15]	$\left(\frac{n}{\varepsilon^2}\right)^{O(\log(\ell)/\varepsilon^2)}$	$\left(\frac{n}{\varepsilon^2}\right)^{O(\log(\ell)/\varepsilon^2)}$	Yes
Folded RS/Univariate Multiplicity Codes [GW13]	$\left(\frac{n\ell}{\varepsilon^2}\right)^{O(\ell/\varepsilon^2)}$	$\left(\frac{n\ell}{\varepsilon}\right)^{O(\ell/\varepsilon)}$	Yes
Folded RS/Univariate Multiplicity Codes [KRZSW18, KRZSW23, Tam24]	$\left(\frac{n\ell}{\varepsilon^2}\right)^{O(\ell/\varepsilon^2)}$	$(\ell/\varepsilon)^{O((1+\log \ell)/\varepsilon)}$	Yes
Folded AG Codes [GX13, GK16]	$\exp\left(\frac{\ell \log(\ell/\varepsilon)}{\varepsilon^2}\right)$	$2^{2^{O_{\varepsilon,\ell}(\log^*(n))}}$	Yes
Codes Based on the AEL Framework [ST25]	$\exp(\exp(\ell/\varepsilon \log(\ell/\varepsilon)))$	$\exp(\exp(\ell/\varepsilon \log(\ell/\varepsilon)))$	Yes
Codes Based on the AEL Framework [JS25]	$\exp((\ell/\varepsilon)^{(\ell/\varepsilon)^{O(\ell/\varepsilon)}})$	$(\ell/\varepsilon)^{O(\ell/\varepsilon)}$	Yes
Folded RS/Univariate Multiplicity Codes (This Work)	$\left(\frac{n\ell}{\varepsilon^2}\right)^{O(\ell/\varepsilon^2)}$	$\left(\frac{\ell}{R+\varepsilon}\right)^{O(R/\varepsilon)}$	Yes
Any Linear Code [GLM ⁺ 22, CZ25, LMS25, LS25]	Any	$\ell^{\Omega(R/\varepsilon)}$	Lower Bound

Table 1: Table 1: Constructions of (α, ℓ, L) list recoverable codes of rate $R = R^* - \varepsilon$, where $R^* = 1 - \alpha$ is list-recovering capacity (when $|\Sigma| \geq (1 + \ell)^{\Omega(1/\varepsilon)}$). We assume that $R^* \in (0, 1)$ is constant (independent of n, ε, ℓ).

4 Mathematical Background

4.1 Basic Coding Theory

Bowen, feel free to add more basic coding theory and list recovery background definitions if you think they are relevant to understanding

Many of the below definitions are directly taken, or at least adapted, from the book *Essential Coding Theory* [GRS25].

Definition 1. A code of block length n over an alphabet Σ is a subset of Σ^n . Typically, we will use q to denote the alphabet size, $|\Sigma|$.

Definition 2. Given a code $C \subseteq \Sigma^n$, we say that its dimension is as follows:

$$k \stackrel{\text{def}}{=} \log_q |C|.$$

Definition 3. The rate of a code with dimension k and block length n is given by

$$R \stackrel{\text{def}}{=} \frac{k}{n}$$

Definition 4. A code $C \subseteq \mathbb{F}_q^n$ is called a linear code if it is a k -dimensional linear subspace of $\mathbb{F}_q^n$ for some $k \leq n$. In this case, C has dimension k and rate $R = \frac{k}{n}$.

Definition 5. [AGG⁺25]. A random linear code of rate R and block length n over $\mathbb{F}_q$ is a random subspace of $\mathbb{F}_q^n$ of dimension Rn .

Definition 6. [Gur10]. For integers $1 \leq k < n$, a field $\mathbb{F}$ of size $|\mathbb{F}| \geq n$, and a set $S = \{\alpha_1, \dots, \alpha_n\} \subseteq \mathbb{F}$, we define the Reed–Solomon code

$$\text{RS}_{\mathbb{F},S}[n, k] = \{(p(\alpha_1), p(\alpha_2), \dots, p(\alpha_n)) \in \mathbb{F}^n \mid p \in \mathbb{F}[X] \text{ is a polynomial of degree } \leq k-1\}.$$

A natural interpretation of the code $\text{RS}_{\mathbb{F},S}[n, k]$ is via its encoding map. To encode a message

$$m = (m_0, m_1, \dots, m_{k-1}) \in \mathbb{F}^k,$$

we interpret the message as the polynomial

$$p(X) = m_0 + m_1X + \dots + m_{k-1}X^{k-1} \in \mathbb{F}[X].$$

We then evaluate the polynomial p at the points $\alpha_1, \alpha_2, \dots, \alpha_n$ to obtain the codeword corresponding to m .

Equivalently, to evaluate the polynomial p on the points $\alpha_1, \alpha_2, \dots, \alpha_n$, we multiply the message vector m on the left by the $n \times k$ Vandermonde matrix

$$G = \begin{pmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{k-1} \\ 1 & \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^{k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \cdots & \alpha_n^{k-1} \end{pmatrix}.$$

The matrix G is a generator matrix for $\text{RS}_{\mathbb{F},S}[n, k]$, and hence Reed–Solomon codes are linear codes over $\mathbb{F}$.

Definition 7. [Gur10] Let $\mathbb{F}$ be a field of size q whose nonzero elements are $\{1, \gamma, \dots, \gamma^{n-1}\}$ for $n = q - 1$. Let $s \geq 1$ be an integer which divides n , and let $1 \leq k < n$ be the degree parameter.

The folded Reed–Solomon code $\text{FRS}_{\mathbb{F}}^{(s)}[k]$ is a code over the alphabet $\mathbb{F}^s$ that encodes a polynomial $f \in \mathbb{F}[X]$ of degree at most $k - 1$ as

$$f(X) \mapsto \left(\begin{bmatrix} f(1) \\ f(\gamma) \\ \vdots \\ f(\gamma^{s-1}) \end{bmatrix}, \begin{bmatrix} f(\gamma^s) \\ f(\gamma^{s+1}) \\ \vdots \\ f(\gamma^{2s-1}) \end{bmatrix}, \dots, \begin{bmatrix} f(\gamma^{n-s}) \\ f(\gamma^{n-s+1}) \\ \vdots \\ f(\gamma^{n-1}) \end{bmatrix} \right). \quad (1)$$

Observe that the folded Reed–Solomon code has block length $N = n/s$ and rate $R = k/n$, which is the same as the original, unfolded Reed–Solomon code.

Remark 1. Folded Reed–Solomon codes are, effectively, nothing more than Reed–Solomon codes over a larger alphabet, but equipped with a special folding operation that groups consecutive evaluations of a polynomial.

This folding operation significantly improves list-decoding and list-recovery performance.

4.2 List Recovery

Definition 8. Formally, a code $\mathcal{C}$ is (ρ, ℓ, L) -list-recoverable if for any sets $S_1, \dots, S_n \subset \mathbb{F}_q$ of size $|S_i| = \ell$, there are at most L codewords $c_1, \dots, c_L \in \mathcal{C}$ such that $c_i \in S_i$ for at least a $(1 - \rho)n$ fraction of the coordinates.

Remark 2. $(\rho, 1, 1)$ list-recoverability is equivalent to ρ unique-decodability.

Remark 3. $(\rho, 1, L)$ list-recoverability is equivalent to (ρ, L) list-decodability.

Theorem 3 (List Recovery Capacity Theorem). [RV25] Let $q := |\Sigma| \geq 2$, let $\ell \in [1, q-1]$, and define the (q, ℓ) -entropy function $h_{q, \ell}$ by

$$h_{q, \ell}(x) := x \log_q \left(\frac{q - \ell}{x} \right) + (1 - x) \log_q \left(\frac{\ell}{1 - x} \right), \quad x \in (0, 1 - \ell/q).$$

Fix $\rho \in (0, 1 - \ell/q)$ and $\varepsilon \in (0, 1 - h_{q, \ell}(\rho))$. Then for all sufficiently large block lengths n , the following hold.

- **(Possibility)** A random code of length n and rate

$$R = 1 - h_{q, \ell}(\rho) - \varepsilon$$

is $(\rho, \ell, O(\ell/\varepsilon))$ list-recoverable with probability at least $1 - q^{-\varepsilon n}$.

- **(Impossibility)** Any code of length n and rate

$$R = 1 - h_{q, \ell}(\rho) + \varepsilon$$

is not $(\rho, \ell, q^{o(\varepsilon n)})$ list-recoverable.

“If a code of rate $R = 1 - h_{q, \ell}(\rho) - \varepsilon$ is (ρ, ℓ, L) list-recoverable with $L \leq \text{poly}(n)$, then we say that the code achieves list recovery capacity. If, additionally, the output list size L can be taken to be $O(\ell/\varepsilon)$, then we say that the code achieves the Elias bound.” [RV25]

It is known that Random Linear Codes achieve list-recovery capacity. However, we do not know if they achieve the Elias bound. [RV25].

The [BCDZ25a] $\rho^* = 0$ case of the Chen-Zhang conjecture can be thought of as a zero-error list-recovery version of the generalized Singleton bound for list-decoding.

One of the biggest motivations to better understand list recovery is its robustness as a tool to better understand list *decoding*. Specifically, it is common to construct *concatenated* codes as a means of obtaining codes with better list-decoding parameters. These concatenated codes consist of an outer and inner code; if the outer code is (ξ, ℓ, L) list-recoverable and the inner code is (ρ, ℓ) list-decodable, then the concatenated code is $(\xi\rho, L)$ list-decodable [RV25].

4.3 Subspace Designs

Intuitively, a subspace design is a collection of subspaces of vector spaces such that they don’t overlap too much. They are well-spread in a low-dimensional direction.

Definition 9 (Subspace Design). Let $H_1, \dots, H_m \subseteq \mathbb{F}_q^k$ be spaces with codimension s . We say that these spaces form a (d, A) -subspace design if for all d -dimensional subspaces $U \subseteq \mathbb{F}_q^k$, we have that

$$\sum_{i=1}^m \dim(U \cap H_i) \leq A.$$

5 Calculations and Attempts

5.1 Introduction

We aim to imitate the approach of [BCDZ25a] and explore whether similar ideas can be adapted to *nonlinear* code settings.

To set the stage, recall the current frontier:

- Random codes near list-recovery capacity can achieve output list size $L = O(\ell/\varepsilon)$.
- In contrast, random *linear* codes near list-recovery capacity may require $L \geq \ell^{\Omega(1/\varepsilon)}$ (see *Bounds for list-decoding and list-recovery of random linear codes* by Guruswami–Li–Mosheiff–Resch–Silas–Wootters, 2020).

More precise statement are also included below

Theorem 4 (Random (nonlinear) codes achieve $L = O(\ell/\varepsilon)$ near capacity). *Fix alphabet size $q := |\Sigma|$, list size $\ell \geq 2$, and $\varepsilon > 0$. A uniformly random (not-necessarily-linear) code of rate*

$$R = 1 - \log_q(\ell) - \varepsilon$$

is, with high probability, $(\rho = 0, \ell, L)$ -list recoverable with

$$L = O\left(\frac{\ell}{\varepsilon}\right).$$

Theorem 5 (Singleton-type lower bounds force $L = \Omega(\ell/\varepsilon)$ near capacity). *In the large-alphabet regime, for rates approaching list-recovery capacity, any (ρ, ℓ, L) -list-recoverable code must have output list size at least*

$$L = \Omega\left(\frac{\ell}{\varepsilon}\right)$$

when the rate is within ε of capacity (up to constant factors depending on the precise model/parameters). Equivalently, the dependence $L = \Theta(\ell/\varepsilon)$ is information-theoretically tight (up to constant factors) for general codes.

These results motivate the following target.

Open Problem 1 (Explicit (possibly nonlinear) codes meeting $L = \Theta(\ell/\varepsilon)$ with efficient recovery). *Construct explicit (deterministically describable) code families, not necessarily linear, that achieve the information-theoretic near-capacity benchmark*

$$L = \Theta\left(\frac{\ell}{\varepsilon}\right)$$

in the standard list-recovery-from-errors model, together with efficient list-recovery algorithms.

This section records preliminary attempts toward this goal.

5.2 Attack Plan

The proof strategy of [BCDZ25a] can be distilled into three steps:

1. Identify a combinatorial object that “spreads” subspaces in a controlled way (in [BCDZ25a], this is a subspace design).
2. Use this object to define a natural code (in [BCDZ25a], the code is the image of linear maps).
3. Apply an entropy inequality to convert the combinatorial condition into a list-recovery guarantee (in [BCDZ25a], via Brascamp–Lieb inequalities).

We follow the same template, but modify the ingredients in each step.

5.3 Attempt 1: Shearer’s Lemma (toy attempt; fails near capacity)

What we tried and why. As a first (and intentionally simple) test case, we replace the Brascamp–Lieb step with Shearer-type inequalities. The hope is that a clean entropy argument might already yield a meaningful list-recovery bound in a more general (possibly nonlinear) setting.

What we learn. This attempt clarifies exactly where Shearer is too weak: it yields a sufficient condition of the form

$$\log(L+1) \lesssim \frac{W(L)}{r(L)} \log \ell,$$

so unless one can make $\frac{r(L)}{W(L)}$ large, the resulting list size is poor. In particular, naively applying Shearer still leads to an exponential-type bound in ℓ , so at best one expects something like $O(\ell^{1+o(1)})$ (still far from the nonlinear near-capacity benchmark).

How this matches the “three steps.”

1. The combinatorial objects are projective spaces.
2. We use the natural projections.
3. The entropy inequality is (fractional) Shearer’s inequality.

5.3.1 Shearer preliminaries

Definition 10 (Entropy). *Let X be a discrete random variable with finite support. Its (base-2) Shannon entropy is*

$$H(X) := - \sum_{x \in (X)} \Pr[X = x] \log \Pr[X = x].$$

If X is uniform on a finite set S , then $H(X) = \log |S|$. If $Y = f(X)$ for a deterministic function f , then $H(Y) \leq H(X)$.

Definition 11 (Projection notation). *Let $U = (U_1, \dots, U_t)$ be a random vector and let $S \subseteq [t]$. Write U_S for the subvector $(U_j)_{j \in S}$.*

Definition 12 (Fractional cover). *Let $\mathcal{S} = \{S_1, \dots, S_n\}$ be subsets of $[t]$ and let $w_1, \dots, w_n \geq 0$. We say that $w = (w_i)$ is a fractional r -cover of $[t]$ by $\mathcal{S}$ if for every $j \in [t]$,*

$$\sum_{i: j \in S_i} w_i \geq r.$$

Write $W := \sum_{i=1}^n w_i$ for the total weight.

Theorem 6 (Fractional Shearer’s inequality). *Let $U = (U_1, \dots, U_t)$ be a random vector, let $\mathcal{S} = \{S_1, \dots, S_n\}$ be subsets of $[t]$, and let $w = (w_i)$ be a fractional r -cover. Then*

$$H(U) \leq \frac{1}{r} \sum_{i=1}^n w_i H(U_{S_i}).$$

Remark 4. *The unweighted Shearer inequality corresponds to $w_i \equiv 1$ and $r = \min_{j \in [t]} |\{i : j \in S_i\}|$.*

5.3.2 List recovery and the Shearer-design hypothesis

Definition 13 ((Zero-error) list recovery). *Let $C \subseteq \Sigma^n$. For integers $\ell, L \geq 1$, we say that C is $(0, \ell, L)$ -list recoverable if for every sequence of sets $T_1, \dots, T_n \subseteq \Sigma$ with $|T_i| \leq \ell$,*

$$\mathcal{L}(T_1, \dots, T_n) := \{c \in C : c_i \in T_i \text{ for all } i \in [n]\}$$

has size $|\mathcal{L}(T_1, \dots, T_n)| \leq L$. Equivalently, there is no set $E \subseteq C$ of size $L + 1$ such that for every i , the set $\{c_i : c \in E\}$ has size at most ℓ .

Definition 14 (Code from maps). *Let $\pi_1, \dots, \pi_n : \mathbb{F}_q^k \rightarrow \mathbb{F}_q^s$ be maps. Define*

$$C := \{(\pi_1(x), \dots, \pi_n(x)) : x \in \mathbb{F}_q^k\} \subseteq (\mathbb{F}_q^s)^n.$$

Definition 15 (Fractional Shearer-design hypothesis up to dimension L). *Fix an integer $L \geq 1$. We say that $\pi_1, \dots, \pi_n$ satisfy the fractional Shearer-design hypothesis up to dimension L with parameters $r(L) > 0$ and $W(L) > 0$ if the following holds.*

For every linear subspace $V \leq \mathbb{F}_q^k$ with $t := \dim(V) \leq L$, there exist:

- *an injective linear map $\phi_V : V \rightarrow \mathbb{F}_q^t$,*
- *subsets $S_1(V), \dots, S_n(V) \subseteq [t]$,*
- *decoders $g_{i,V} : \mathbb{F}_q^s \rightarrow \mathbb{F}_q^{S_i(V)}$ for $i \in [n]$,*
- *nonnegative weights $w_1^{(V)}, \dots, w_n^{(V)} \geq 0$,*

such that:

1. *(local recoverability) for all $x \in V$ and $i \in [n]$,*

$$g_{i,V}(\pi_i(x)) = \phi_V(x)_{S_i(V)}.$$

2. *(fractional cover) for all $j \in [t]$,*

$$\sum_{i: j \in S_i(V)} w_i^{(V)} \geq r(L).$$

3. *(total weight)*

$$\sum_{i=1}^n w_i^{(V)} \leq W(L).$$

5.3.3 Fractional Shearer analogue of Theorem 3.1

Theorem 7 (Fractional Shearer analogue of Theorem 3.1 (zero-error)). *Let $C = \{(\pi_1(x), \dots, \pi_n(x)) : x \in \mathbb{F}_q^k\}$. Fix integers $L \geq \ell \geq 2$. Assume that the maps $\pi_1, \dots, \pi_n$ satisfy Definition 15 (fractional Shearer-design hypothesis) up to dimension L with parameters $r(L)$ and $W(L)$. Then C is $(0, \ell, L)$ -list recoverable provided*

$$\log(L+1) > \frac{W(L)}{r(L)} \log \ell.$$

Equivalently,

$$\frac{r(L)}{W(L)} > \frac{\log \ell}{\log(L+1)}.$$

Proof. Assume for contradiction that C is not $(0, \ell, L)$ -list recoverable. Then there exists $E \subseteq C$ with $|E| = L+1$ such that for every $i \in [n]$,

$$|\{c_i : c \in E\}| \leq \ell.$$

Let $F \subseteq \mathbb{F}_q^k$ be the preimage of E under $x \mapsto (\pi_1(x), \dots, \pi_n(x))$. Then $|F| = |E| = L+1$. Translating F if needed, assume $0 \in F$ (this does not change $|\pi_i(F)|$).

Let $V := \text{span}(F)$ and $t := \dim(V) \leq L$. Let X be uniform on F , so

$$H(X) = \log |F| = \log(L+1).$$

Apply Definition 15 to V : let $\phi_V : V \rightarrow \mathbb{F}_q^t$ be injective and set $U := \phi_V(X)$. Injectivity on F gives

$$H(U) = H(X) = \log(L+1).$$

For each i , the decoder satisfies $g_{i,V}(\pi_i(x)) = \phi_V(x)_{S_i(V)}$ for all $x \in V$, hence

$$U_{S_i(V)} = g_{i,V}(\pi_i(X)), \quad \text{so} \quad H(U_{S_i(V)}) \leq H(\pi_i(X)).$$

Using fractional Shearer (Theorem 6) with weights $w_i^{(V)}$,

$$H(U) \leq \frac{1}{r(L)} \sum_{i=1}^n w_i^{(V)} H(U_{S_i(V)}) \leq \frac{1}{r(L)} \sum_{i=1}^n w_i^{(V)} H(\pi_i(X)).$$

Since $\pi_i(X)$ is supported on $\pi_i(F)$ and $|\pi_i(F)| \leq \ell$, we have $H(\pi_i(X)) \leq \log \ell$. Therefore,

$$\log(L+1) = H(U) \leq \frac{1}{r(L)} \left(\sum_{i=1}^n w_i^{(V)} \right) \log \ell \leq \frac{W(L)}{r(L)} \log \ell,$$

contradicting $\log(L+1) > \frac{W(L)}{r(L)} \log \ell$. Thus no such E exists, and C is $(0, \ell, L)$ -list recoverable. $\square$

5.3.4 A Corollary 3.2 analogue

Corollary 1 (Shearer analogue of Corollary 3.2). *Fix $\ell \geq 2$, and let $R \in (0, 1)$ and $\varepsilon > 0$. Let $C = \{(\pi_1(x), \dots, \pi_n(x)) : x \in \mathbb{F}_q^k\}$ be as above. Define*

$$L := \left\lfloor \ell^{1/(1-R-2\varepsilon)} \right\rfloor.$$

Assume that $\pi_1, \dots, \pi_n$ satisfy the fractional Shearer-design hypothesis (Definition 15) up to dimension L with parameters $r(L), W(L)$ such that

$$\frac{r(L)}{W(L)} \geq 1 - R - \varepsilon.$$

Then for all sufficiently large ℓ (equivalently, for L sufficiently large), the code C is $(0, \ell, L)$ -list recoverable.

Proof. By the choice of L , for sufficiently large ℓ we have $L + 1 \leq 2\ell^{1/(1-R-2\varepsilon)}$, hence

$$\log(L + 1) \leq \log 2 + \frac{1}{1 - R - 2\varepsilon} \log \ell.$$

For large ℓ , the $\log 2$ term is negligible relative to $\log \ell$, so

$$\frac{\log \ell}{\log(L + 1)} \leq 1 - R - 2\varepsilon.$$

Therefore,

$$\frac{\log \ell}{\log(L + 1)} \leq 1 - R - 2\varepsilon < 1 - R - \varepsilon \leq \frac{r(L)}{W(L)}.$$

This matches the sufficient condition in Theorem 7, so C is $(0, \ell, L)$ -list recoverable. $\square$

5.4 Attempt 2: Double-counting and finite geometry (works for L , fails for rate)

What we tried and why. Attempt 1 showed that a direct entropy approach via Shearer can force poor list-size bounds unless one has a very strong “cover ratio” $\frac{r(L)}{W(L)}$. In this attempt we switch tactics: instead of bounding entropy, we bound the number of consistent codewords by a direct *incidence double-counting* argument. The hope is that in very structured geometries, the replication of points across blocks is uniform and yields an $L = O(\ell/\varepsilon)$ -type dependence essentially for free.

What we learn. The combinatorics here is strong enough to give the *right* list-size scaling (linear in ℓ). The bottleneck is instead the *rate*: the most naive finite-geometry incidence codes have vanishing rate as soon as one grows q or n in a meaningful way.

How this matches the “three steps.”

1. The combinatorial objects are affine spaces (incidence structures of points and hyperplanes).
2. The encoding uses the natural “projection”/membership test: either reveal the point (if it lies in a block) or output a special symbol.
3. The analysis is by a clean double-counting bound using the replication number.

Rate of the naive construction. Here $|C| = |\Omega| = q^m$ and $|\Sigma| = |\Omega| + 1 = q^m + 1$. With $n = tq$, the (block) rate is

$$R = \frac{\log |C|}{n \log |\Sigma|} \approx \frac{m \log q}{(tq) \cdot m \log q} = \frac{1}{tq}.$$

Thus $R \approx 1/(tq)$, i.e. it vanishes as soon as q grows (or as soon as one wants n to grow in any meaningful way). So while the list-size bound is optimal, the construction is far from near-capacity.

5.4.1 Incidence codes and a replication-based list-size bound

Definition 16 (Incidence system with bounded replication). *Let Ω be a finite set of “points” and let $\mathcal{B} = \{B_1, \dots, B_n\}$ be a family of subsets $B_i \subseteq \Omega$ (“blocks”). Define the replication numbers*

$$r_{\min} := \min_{x \in \Omega} |\{i : x \in B_i\}|, \quad r_{\max} := \max_{x \in \Omega} |\{i : x \in B_i\}|.$$

Definition 17 (Incidence code). Fix q and identify Ω with a subset of $\mathbb{F}_q^m$ (or treat Ω as an abstract alphabet). Let $\perp$ be a special symbol not in Ω and set $\Sigma := \Omega \cup \{\perp\}$. Given an incidence system $(\Omega, \mathcal{B})$, define maps $\pi_i : \Omega \rightarrow \Sigma$ by

$$\pi_i(x) := \begin{cases} x, & x \in B_i, \\ \perp, & x \notin B_i. \end{cases}$$

Define the code

$$C(\Omega, \mathcal{B}) := \{(\pi_1(x), \dots, \pi_n(x)) : x \in \Omega\} \subseteq \Sigma^n.$$

Theorem 8 (Incidence-code list recovery via replication). Let $(\Omega, \mathcal{B})$ be an incidence system with blocks $\mathcal{B} = \{B_1, \dots, B_n\}$ and replication number $r_{\min}$. Then the incidence code $C(\Omega, \mathcal{B})$ is $(0, \ell, L)$ -list recoverable with

$$L = \left\lfloor \frac{n\ell}{r_{\min}} \right\rfloor.$$

In particular, for every choice of lists $T_1, \dots, T_n \subseteq \Sigma$ with $|T_i| \leq \ell$,

$$|\mathcal{L}(T_1, \dots, T_n)| \leq \frac{n\ell}{r_{\min}}.$$

Proof. Fix lists $T_1, \dots, T_n \subseteq \Sigma$ with $|T_i| \leq \ell$. Let $S := \mathcal{L}(T_1, \dots, T_n) \subseteq C(\Omega, \mathcal{B})$ be the consistent codewords, and define the corresponding set of points

$$X := \{x \in \Omega : (\pi_1(x), \dots, \pi_n(x)) \in S\}.$$

Clearly $|S| = |X|$.

For each $i \in [n]$, define the set of *allowed points inside block B_i* :

$$A_i := B_i \cap (T_i \setminus \{\perp\}) \subseteq \Omega.$$

Since $|T_i| \leq \ell$, we have $|A_i| \leq \ell$ for every i .

Now fix $x \in X$. For each i with $x \in B_i$, we have $\pi_i(x) = x$, and consistency implies $x \in T_i$, hence $x \in A_i$. Therefore x lies in at least $|\{i : x \in B_i\}| \geq r_{\min}$ of the sets A_i . Double-counting incidences gives

$$r_{\min} |X| \leq \sum_{i=1}^n |A_i| \leq \sum_{i=1}^n \ell = n\ell,$$

so $|X| \leq n\ell/r_{\min}$, i.e. $|S| \leq n\ell/r_{\min}$, as claimed. $\square$

5.4.2 Explicit incidence structure from higher-dimensional affine geometry

Theorem 8 reduces list recovery to a single combinatorial parameter: the *replication number* $r_{\min}$, i.e. the minimum number of blocks that contain any given point. To get an explicit family with large (and ideally uniform) replication, it is natural to use finite geometry, where incidence patterns are highly regular.

A standard source of such regularity is the affine space (m, q) , whose points are $\mathbb{F}_q^m$ and whose affine hyperplanes are the solutions to a single nontrivial linear equation $a(x) = b$. Fixing a linear functional a gives a whole *parallel class* of hyperplanes $\{a(x) = b : b \in \mathbb{F}_q\}$, which partitions $\mathbb{F}_q^m$ into q disjoint hyperplanes, each of size q^{m-1} . We will take t such parallel classes corresponding to t distinct directions.

Definition 18 (Affine hyperplane parallel classes in $\mathbb{F}_q^m$). Fix $m \geq 2$ and a prime power q , and let $\Omega := \mathbb{F}_q^m$.

Choose nonzero linear functionals $a_1, \dots, a_t \in (\mathbb{F}_q^m)^*$ such that no two are proportional. Equivalently, the kernels $\ker(a_j)$ are t distinct codimension-1 linear subspaces, so the equations $a_j(x) = b$ represent t distinct hyperplane directions.

For each $j \in [t]$ and each $b \in \mathbb{F}_q$, define the affine hyperplane

$$H_{j,b} := \{x \in \mathbb{F}_q^m : a_j(x) = b\}.$$

Let

$$\mathcal{B} := \{H_{j,b} : j \in [t], b \in \mathbb{F}_q\}, \quad n := |\mathcal{B}| = tq.$$

Remark 5 (Basic structure of the blocks). Fix a direction j . As b ranges over $\mathbb{F}_q$, the sets $H_{j,b}$ are q pairwise disjoint hyperplanes whose union is all of $\mathbb{F}_q^m$:

$$\mathbb{F}_q^m = \bigsqcup_{b \in \mathbb{F}_q} H_{j,b}.$$

Indeed, every $x \in \mathbb{F}_q^m$ has a unique value $b = a_j(x)$, so it belongs to exactly one hyperplane in that parallel class. Moreover, each hyperplane has size $|H_{j,b}| = q^{m-1}$ since it is an affine translate of $\ker(a_j)$.

Lemma 1 (Replication numbers in the affine construction). In the above construction, every point $x \in \mathbb{F}_q^m$ lies in exactly one hyperplane $H_{j,b}$ for each fixed j , hence lies in exactly t blocks overall. Therefore

$$r_{\min} = r_{\max} = t.$$

Proof. Fix $x \in \mathbb{F}_q^m$ and a direction j . The value $b = a_j(x) \in \mathbb{F}_q$ is uniquely determined, and by definition $x \in H_{j,b}$. If $b' \neq b$, then $a_j(x) = b \neq b'$, so $x \notin H_{j,b'}$. Thus, for each fixed j , the point x lies in exactly one block among $\{H_{j,b}\}_{b \in \mathbb{F}_q}$.

Since there are t choices of j , the point x lies in exactly t blocks in total. This shows that $|\{i : x \in B_i\}| = t$ for every x , hence $r_{\min} = r_{\max} = t$. $\square$

Corollary 2 (Explicit $(0, \ell, L)$ -list recovery from (m, q) hyperplanes). Let $m \geq 2$, q a prime power, and choose any $t \geq 1$ distinct hyperplane directions as above, giving $n = tq$ coordinates. Then the incidence code $C(\Omega, \mathcal{B}) \subseteq (\Omega \cup \{\perp\})^n$ is $(0, \ell, L)$ -list recoverable with

$$L = \left\lfloor \frac{n\ell}{t} \right\rfloor = \lfloor q\ell \rfloor.$$

The alphabet size is $|\Sigma| = |\Omega| + 1 = q^m + 1$.

Proof. By Lemma 1, the replication number is $r_{\min} = t$, and the number of blocks is $n = tq$. Applying Theorem 8 yields

$$L = \left\lfloor \frac{n\ell}{r_{\min}} \right\rfloor = \left\lfloor \frac{tq\ell}{t} \right\rfloor = \lfloor q\ell \rfloor.$$

$\square$

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